Huawei Access Node KR-ACC-02 QoS Congestion Event (ALARM_7946) Warning Resolution SOP

Node: KR-ACC-02 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_7946

Description

The Huawei Access Node KR-ACC-02 at KR Puram Access site has reported a QoS Congestion Event (Microbursts, interface tail drops, QoS queue starvation) warning alarm (ALARM_7946). This failure domain occurs due to the inherent design of the Huawei NE40E1 access node's QoS scheduling mechanism, which may cause queue starvation in certain network conditions. When the available bandwidth of the interface is insufficient to handle the incoming traffic, the QoS scheduler may drop packets, leading to microbursts and tail drops. This can cause the QoS queue to become congested, resulting in the warning alarm. Additionally, the Huawei access node's hardware implementation of the QoS scheduling algorithm may also contribute to the congestion event. A deeper analysis of the traffic patterns and QoS configuration is required to identify the root cause of the congestion event.

Pre-requisite Checks

- Verify the power supply to the access node is stable and within normal operating parameters.

- Check the redundant link status to ensure it is functioning correctly and not contributing to the congestion event.

- Dispatch remote hands to the KR Puram Access site to perform physical checks on the access node and its surroundings.

- Collect traffic statistics and QoS configuration data from the access node using the Huawei CLI.

Potential Root Causes

Cause	Probability
Insufficient interface bandwidth	High
Misconfigured QoS policies	Medium
Hardware implementation issues with QoS scheduling algorithm	Low

Diagnostic Commands & Expected Outputs

```
Command: display qos queue statistics, Expected Output: Queue statistics for all queues on the interface, Description: This command displays the queue statistics for all queues on the interface, including the number of packets dropped and the queue depth. This information can help identify if the QoS queue is congested and if so, which queue is experiencing the congestion.

Command: display interface queue statistics, Expected Output: Interface queue statistics, including packet drop count and queue
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```
depth, Description: This command displays the interface queue statistics, including the number of packets dropped and the queue depth. This information can help identify if the interface is experiencing congestion and if so, which queue is experiencing the congestion.

Command: display qos policy, Expected Output: QoS policy configuration for the interface, Description: This command displays the QoS policy configuration for the interface, including the QoS policies applied to the interface and the priority of each policy. This information can help identify if the QoS policies are misconfigured and if so, which policy is causing the congestion.
```

Resolution Steps

Step 1: Step: Adjust QoS policies to prioritize critical traffic, Command: modify qos policy priority , Description: This command adjusts the priority of the QoS policy to prioritize critical traffic. By prioritizing critical traffic, the QoS scheduler can ensure that critical traffic is delivered to the network without delay.

Step 2: Step: Increase interface bandwidth, Command: modify interface bandwidth , Description: This command increases the bandwidth of the interface to ensure that it can handle the incoming traffic without congestion. By increasing the bandwidth, the interface can deliver traffic at a higher rate, reducing the likelihood of congestion events.

Step 3: Step: Reset QoS scheduler, Command: reset qos scheduler, Description: This command resets the QoS scheduler to its default state. By resetting the QoS scheduler, any misconfigured QoS policies or hardware implementation issues can be resolved, ensuring that the QoS scheduler functions correctly and does not contribute to congestion events.

Post-Check Verification

```
Command: display qos queue statistics, Expected Output: Queue statistics for all queues on the interface, Description: This command verifies that the QoS queue is no longer congested and that packets are being delivered to the network without delay.

Command: display interface queue statistics, Expected Output: Interface queue statistics, including packet drop count and queue depth, Description: This command verifies that the interface is no longer experiencing congestion and that packets are being delivered to the network without delay.

Command: display qos policy, Expected Output: QoS policy configuration for the interface, Description: This command verifies that the QoS policies are correctly configured and prioritizing critical traffic.
```

Roll-back Procedures

- Step: Restore original QoS policy configuration, Command: undo modify qos policy, Description: This command restores the original QoS policy configuration, ensuring that any changes made during the troubleshooting process are reversed and the network returns to its original state.

- Step: Restore original interface bandwidth, Command: undo modify interface bandwidth, Description: This command restores the original bandwidth of the interface, ensuring that any changes made during the troubleshooting process are reversed and the network returns to its original state.

- Step: Reset QoS scheduler to original state, Command: undo reset qos scheduler, Description: This command resets the QoS scheduler to its original state, ensuring that any changes made during the troubleshooting process are reversed and the network returns to its original state.